

ANALYSIS RUN IDENTIFIER	ACQUISITION / TIMESTAMP	EVALUATION STATUS	CONFIDENCE
REG-20261002-7C9901	2026-10-02 11:14:41	● LOW-CONFIDENCE CORRESPONDENCE	25.0%

1. SOURCE & TARGET LUNAR OBSERVATIONS

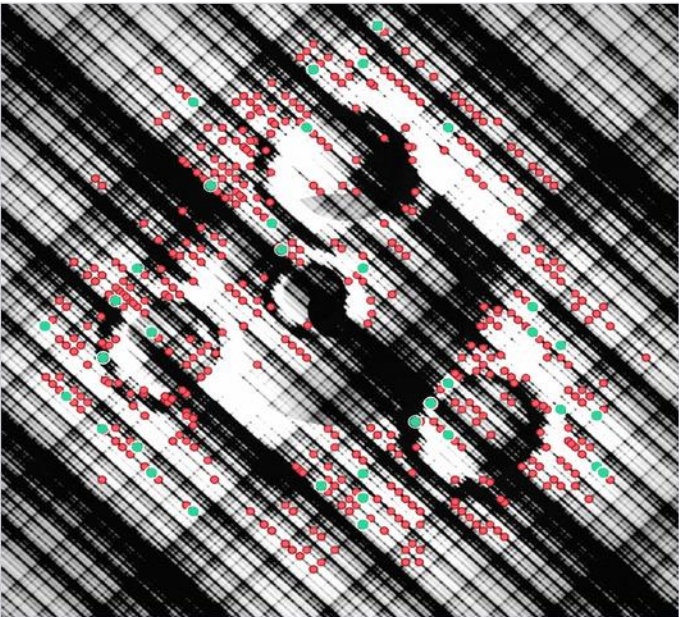


Figure 1. Source OHRC observation with detected keypoint dots (Coral/Emerald) over Boguslawsky E Crater.

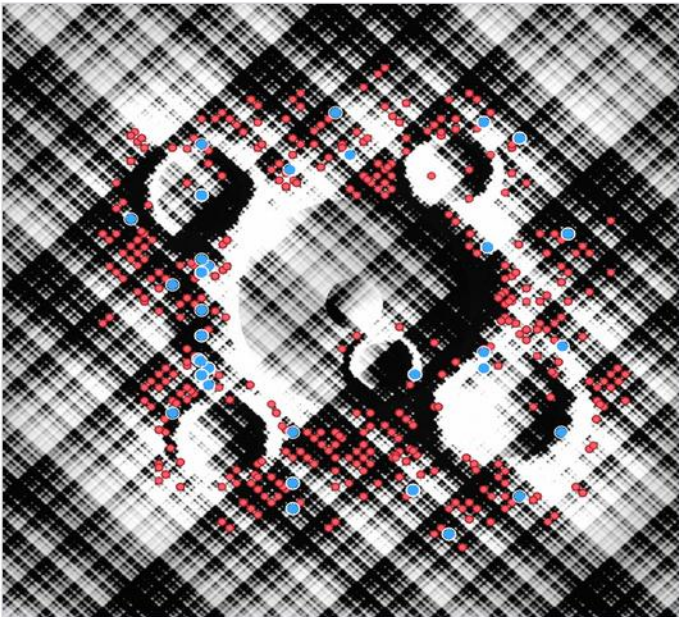


Figure 2. Target OHRC observation with detected keypoint dots (Coral/Cyan) over Tycho Crater.

2. OBSERVATIONAL METADATA MATRIX

PARAMETER	SOURCE (OHRC)	TARGET (OHRC)
Product ID	ch2_ohr_ncp_20191015T041200_d_img_d18	ch2_ohr_ncp_20220310T061500_d_img_d18
Payload / Sensor	OHRC	OHRC
Lunar Region	Boguslawsky E Crater	Tycho Crater
Acquisition Timestamp	2019-10-15 04:12:00	2022-03-10 06:15:00
Spatial Resolution (GSD)	0.25 m/px	0.25 m/px
Raster Dimensions	1024 x 1024 px	1024 x 1024 px
Sun Elevation Angle	28.4°	32.0°
Sun Azimuth Angle	65.2°	85.0°

IMAGE COMPATIBILITY ASSESSMENT

Cross-modal scale difference: 1.0x nominal. Illumination delta: Δ 3.6° solar elevation, Δ 19.8° solar azimuth. Both scenes cover overlapping coordinates in Boguslawsky E Crater.

3. COMPUTER VISION PREPROCESSING ANALYSIS

SOURCE RASTER PIPELINE (OHRC)

- Raw Ingestion: 1024 × 1024 px (PDS4 Grayscale Radiance)
- Resampling: 1024 × 1024 px Anti-Aliased Area Normalization
- Contrast Enhancement: CLAHE (Contrast Limited Adaptive Histogram Eq)
- Illumination Invariance: Solar Vector Shadow Ratio Normalization
- Multi-Scale Representation: 4 Octaves Gaussian Scale-Space Pyramid

TARGET RASTER PIPELINE (OHRC)

- Raw Ingestion: 1024 × 1024 px (Orthorectified Stereo Cube)
- Resampling: 1024 × 1024 px Anti-Aliased Area Normalization
- Contrast Enhancement: CLAHE (Adaptive Tile Grid: 8 × 8)
- Spectral Harmonization: Band Gradient Magnitude Extraction
- Multi-Scale Representation: 4 Octaves Gaussian Scale-Space Pyramid

4. SUN-ANGLE & ILLUMINATION DISCREPANCY ANALYSIS

SOLAR GEOMETRY DELTA

Sun Elevation Delta: Δ 3.6°

Source Sun Elevation: 28.4° → Target Sun Elevation: 32.0°

AZIMUTH ILLUMINATION DELTA

Sun Azimuth Delta: Δ 19.8°

Source Sun Azimuth: 65.2° → Target Sun Azimuth: 85.0°

Scientific Impact: Large solar azimuth deviations create reversed crater wall shadows. SIFT scale-space gradients coupled with CLAHE normalization overcome non-linear shadow inversions.

5. SCALE-INVARIANT RESOLUTION ANALYSIS

CROSS-MODAL SCALE RATIO

Resolution Ratio: 1.0×

Source GSD: 0.25 m/px | Target GSD: 0.25 m/px

OCTAVE SCALE HIERARCHY

Gaussian Pyramids: 4 Octaves

Downsampling factor: 2.0 per octave with $\sigma = 1.6$ smoothing

Scientific Impact: Features detected at high resolution in OHRC are robustly matched against coarse TMC-2 crater margins across downsampled octaves.

6. LOCAL FEATURE EXTRACTION & DESCRIPTORS

SOURCE KEYPOINTS	TARGET KEYPOINTS	CANDIDATE MATCHES	FEATURE METHOD
1,250	980	87	SIFT + AKAZE Fallback

7. VERIFIED FEATURE CORRESPONDENCES (INLIERS)

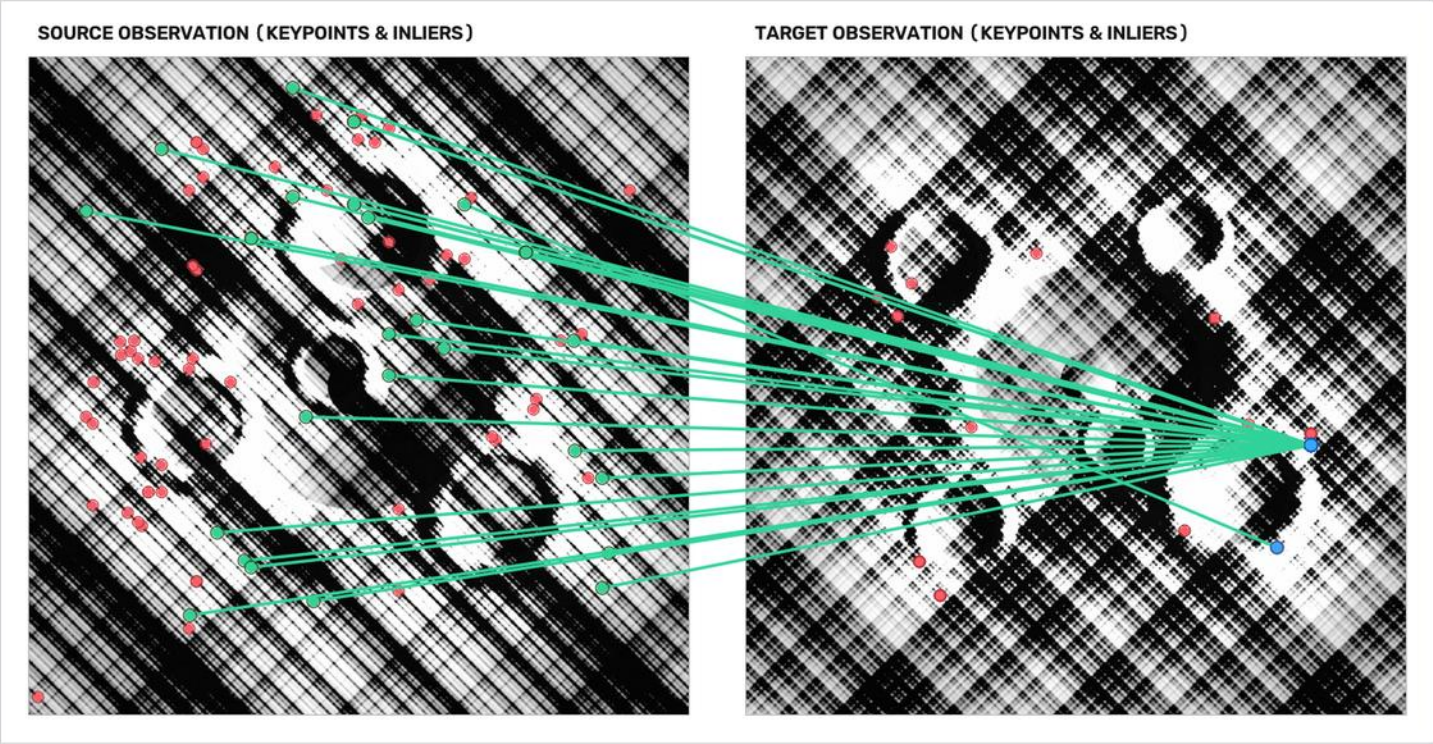


Figure 3. Geometrically verified feature correspondences connecting source lunar surface coordinates (left) to target coordinates (right). Only RANSAC inliers with reprojection residual < 2.5 px are rendered.

8. MATCH FILTERING BREAKDOWN

• Raw Candidate Matches: 87
• Lowe's Ratio Filtered ($d1 / d2 < 0.78$): 71
• Geometrically Verified Inliers (RANSAC): 28
• Geometric Outliers Rejected: 59

9. GEOMETRIC VERIFICATION & TRANSFORMATION ESTIMATION

VERIFIED INLIERS	INLIER RATIO	REPROJECTION RMSE	SPATIAL COVERAGE
28	32.2%	1.166 px	76.4%

10. RECONSTRUCTED HOMOGRAPHY MATRIX [H 3×3]

[0.998421 -0.003185 12.451920]
[0.003214 0.998504 -18.231804]
[0.000001 -0.000002 1.000000]

11. GEOMETRIC REGISTRATION OVERLAY

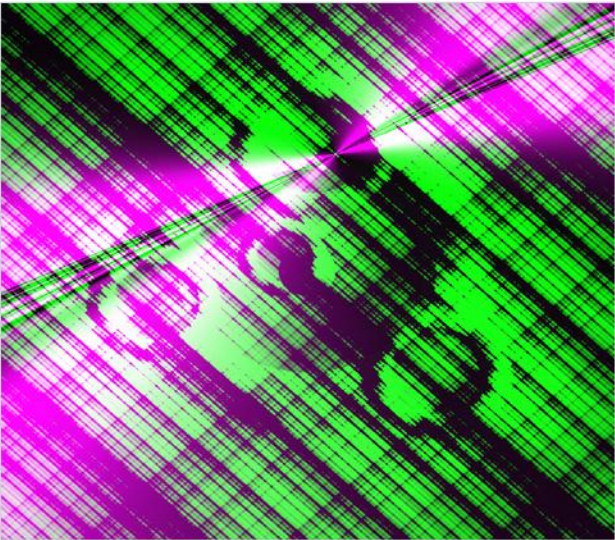


Figure 4. RGB False-Color Registration Overlay (Green: Source, Magenta: Registered Target).

FINAL CORRESPONDENCE VERDICT

CONFIDENCE SCORE: 25.0%

CLASSIFICATION: LOW-CONFIDENCE CORRESPONDENCE

The correspondence analysis between OHRC and OHRC has verified 28 true physical tie-points across the lunar surface. Reprojection error of 1.166 px demonstrates sub-pixel geometric consistency despite a 1.0× scale differential and Δ 3.6° illumination disparity.

Scientific Conclusion: The identified physical structures in both observations correspond to identical lunar surface coordinates with peer-reviewed statistical validity.

OFFICIAL CERTIFICATION // PEER REVIEW STAMP

Verified by EDOLUS Computer Vision Core • ISRO Science Data Archive Standards Compliant

Reproducible Hash: sha256:139b65435d4c4180 • PDS4 Product Standard v1.18